Admissions Committee
Master of Science in Telecommunication Engineering
Politecnico di Milano

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Dear Members of the Admissions Committee,

True engineering is defined not only by what we are taught in the classroom, but by what we choose to build once the lectures end. During my Computer Engineering studies at Politecnico di Torino, I developed a solid foundation in software engineering, computer architecture, and digital systems. While these subjects taught me how computers operate individually, I became increasingly fascinated by the invisible infrastructure that allows millions of systems to function collectively. I realized that the Internet itself is one of the largest and most complex engineering achievements ever created—a continuously evolving distributed system whose reliability depends on careful design rather than chance. Motivated by this realization, I began independently studying enterprise networking, completing the first volume of the official Cisco CCNA guide and currently progressing through the second while simultaneously applying every concept through practical implementation.

For me, understanding networking has never meant memorizing protocols; it has meant discovering why those protocols exist and how they interact to solve real engineering problems. To explore the principles of redundancy, scalability, and fault tolerance, I designed a multi-node Layer 3 Enterprise Campus Network that implements RPVST+, LACP EtherChannels, and secure access-layer protections including PortFast, BPDU Guard, and Root Guard. Building this project taught me that engineering is fundamentally a discipline of balancing resilience, performance, and operational simplicity. Every design decision introduced trade-offs that could only be appreciated through implementation rather than theory alone.

Wanting to understand how enterprise networks securely interact with the public Internet, I expanded my work into an Enterprise Edge Network architecture built around a Defense-in-Depth security model. The design combines OSPF for internal routing, eBGP for external connectivity, and a segmented DMZ (Demilitarized Zone) protected through Static NAT and carefully designed Access Control Lists. Developing this architecture strengthened my understanding that modern networking extends far beyond connectivity. It requires designing systems that remain secure, available, and maintainable even as complexity increases.

As my projects became larger, I encountered another challenge: manual configuration does not scale. This naturally led me toward network programmability and automation. I am currently developing a Python-based automation framework using Netmiko to automate configuration management and verify security compliance across a simulated enterprise WAN migration. Through this work, I have come to appreciate how software engineering and networking are becoming increasingly inseparable. Rather than viewing automation as merely replacing manual tasks, I see it as enabling engineers to build infrastructures that are more consistent, reliable, and adaptable to future demands. Documenting these projects on my public GitHub portfolio and personal website has also become an important part of my learning process, encouraging me to communicate engineering decisions clearly and continuously refine my work.

This convergence of software engineering, distributed systems, and intelligent network infrastructure is precisely why I am applying to the Internet Engineering (Z2D) track at Politecnico di Milano. My background in Computer Engineering allows me to approach networking from both a software and systems perspective, making the interdisciplinary nature of the program particularly attractive. I am especially interested in studying how modern networks evolve through automation, programmability, and software-defined architectures. Courses such as Network Design and Network Automation align directly with the direction in which I have independently developed my skills while providing the theoretical depth necessary to understand not only how these technologies function today, but how they will continue to evolve.

Beyond expanding my technical knowledge, I hope to deepen my understanding of Internet engineering through collaboration with faculty and peers who share the same passion for distributed systems and modern network infrastructure. Graduate study at Politecnico di Milano represents the opportunity to transform years of self-directed learning into rigorous academic expertise while continuing to grow as both an engineer and a lifelong learner.

My long-term ambition is to contribute to the design and automation of resilient large-scale network infrastructures that support cloud platforms, distributed services, and future Internet technologies. I believe Politecnico di Torino has provided me with the analytical foundation necessary to pursue this goal, while my independent projects have strengthened my ability to learn, experiment, and persevere beyond formal instruction. I am confident that the Internet Engineering program at Politecnico di Milano is the ideal environment in which to continue this journey, and I would be honored to contribute my technical curiosity, initiative, and commitment to the program.

Thank you for your time and consideration.

Sincerely,

Resat Orhun Konak